

WASP-12b

R-0576

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231209 · created 2026-07-24T17:22:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460286.9511 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.004 +/- 0.044
Transit depth (Rp/Rs)^2	0.0013 +/- 0.032 [%]
Orbital Inclination (inc)	80.29 +/- 2.94
Airmass coefficient 1 (a1)	1.15 +/- 0.62
Airmass coefficient 2 (a2)	0.03 +/- 0.28
Scatter in the residuals of the lightcurve fit is	55.71 %
Best Comparison Star	#1 - [358, 283]
Optimal Aperture	3.0
Optimal Annulus	5.5
Transit Duration (day)	0.104 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.004 ± 0.044 vs 0.1178 (deviation 2.03σ)
Duration fit vs published	0.104 d vs 0.125 d (deviation 1.37σ)
Mid-time O–C	-1572.0 min
Depth SNR	0.1
Residual scatter	55.71 %

Light curve

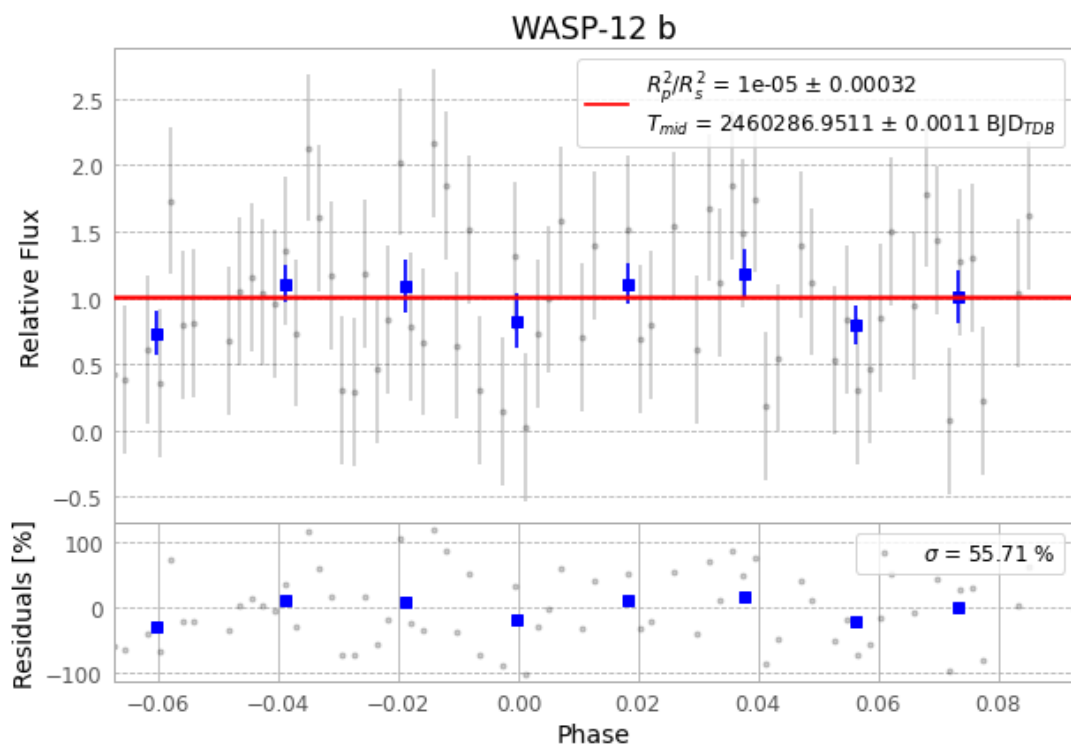


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-08.png (SHA-256 60dbac18caa5611e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 291], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4da2b04f6ae1a998...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

88 FITS frames (58 MB), WASP-12231208085415.FITS → WASP-12231208131515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231209.log (f0ea2a8823f3...), FinalLightCurve_WASP-12 b_2023-12-08.png (60dbac18caa5...), FinalLightCurve_WASP-12 b_2023-12-08.csv (2b58755e6cb7...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 17:22Z.